Use the image from the next page for opcode, funct3 and other values.

1. Add X12, X13, X15;
    a. Convert the given instruction to its corresponding machine code.
    b. Now from the 32 bit binary number you got in question 1.a try to find the instruction itself again.

2. SD X12, 16(X11);
    - Convert the given instruction to its corresponding machine code.
    - Now from the 32 bit binary number you got in question 2.a try to find the instruction itself again.

3. Sll X12, X13, X14;
    - Convert the given instruction to its corresponding machine code.
    - Now from the 32 bit binary number you got in question 3.a try to find the instruction itself again.

4. Slli X12, X16, 62;
    - Convert the given instruction to its corresponding machine code.
    - Now from the 32 bit binary number you got in question 4.a try to find the instruction itself again.

5. a = 5b + 3c + (d/4);

a -> X21

b -> X22

c -> X23

d -> X24

Convert the given code to risc-v assembly instruction(s)

6. a = A[b] + 5;

a -> X21

b -> X22

Base address of the double word array A is in X23

7. Ld X12, 64(X16) ;
    - Convert the given instruction to its corresponding machine code.
    - Now from the 32 bit binary number you got in question 7.a try to find the instruction itself again.

Instruction	Opcode	Funct3	Funct6/7
add	0110011	000	0000000
sub	0110011	000	0100000
sll	0110011	001	0000000
xor	0110011	100	0000000
srl	0110011	101	0000000
sra	0110011	101	0000000
or	0110011	110	0000000
and	0110011	111	0000000
lr.d	0110011	011	0001000
sc.d	0110011	011	0001100
lb	0000011	000	n.a.
lh	0000011	001	n.a.
lw	0000011	010	n.a.
lbu	0000011	100	n.a.
lhu	0000011	101	n.a.
addi	0010011	000	n.a.
slli	0010011	001	000000
xori	0010011	100	n.a.
srli	0010011	101	000000
srai	0010011	101	010000
ori	0010011	110	n.a.
andi	0010011	111	n.a.
jalr	1100111	000	n.a.
sb	0100011	000	n.a.
sh	0100011	001	n.a.
sw	0100011	010	n.a.
beq	1100111	000	n.a.
bne	1100111	001	n.a.
blt	1100111	100	n.a.
bge	1100111	101	n.a.
bltu	1100111	110	n.a.
bgeu	1100111	111	n.a.
lui	0110111	n.a.	n.a.
jal	1101111	n.a.	n.a.